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Simulation of prompt injection attacks on generative pre-trained transformers models

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Abstract

Large language models (LLMs), such as GPT-2, GPT-3.5, and GPT-4o Mini, are widely used in various applications, yet their growing adoption has also revealed critical security vulnerabilities. Among the most pressing threats are prompt injection attacks, sensitive information disclosure, and unbounded consumption, which can lead to model manipulation, data leakage, and denial of service (DoS) conditions. Although prior research has examined these risks individually, this study uniquely conducts a comprehensive side-by-side evaluation of all three adversarial vectors in multiple LLM architectures. Through an automated attack simulation framework, we systematically assess the weaknesses of different models, uncovering notable security gaps. Our results show that newer, more advanced models, such as GPT-4o Mini, are paradoxically more susceptible to adversarial manipulation, achieving a 100% success rate in prompt injection attacks on GPT-4o Mini, compared to up to 90% on GPT-3.5 and 0% on GPT-2. We also analyze how encoding techniques such as Base64, Hex, and Rot-13 affect attack success rates. In sensitive information disclosure tests, GPT-4o Mini achieved 78.57% accuracy with an average similarity score of 0.8401, while GPT-2 still leaked data with 75% accuracy. Furthermore, this research quantifies how unbounded consumption attacks significantly degrade system performance, posing risks to real-world AI deployments. Under simulated unbounded consumption attacks, the response time of the GPT-2 increased from 94.68 seconds at low concurrency to more than 35,000 seconds at 10,000 concurrent requests, indicating extreme vulnerability to DoS exploitation. By bridging the gap between theoretical vulnerabilities and practical adversarial simulations, this study provides actionable insights into AI security. Our findings emphasize the urgent need for robust adversarial defenses, real-time anomaly detection, and privacy-preserving mechanisms to ensure the safe and ethical deployment of LLMs in high-risk environments. These experimental findings challenge the assumption that larger models are more secure and emphasize the importance of embedding robust safeguards in LLM deployments.

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Keywords: Large language models (LLMs); model vulnerabilities; prompt injection attacks; data set privacy; AI security; cyber security

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1. Introduction

Large language models (LLMs), such as GPT-2 and GPT-3.5, are used every day to power applications in a variety of industries, ranging from conversational agents to content generation. However, their increasing popularity has revealed several security flaws that jeopardize their dependability, ethical application, and operational efficiency. Security concerns in LLMs have been recognized, with OWASP identifying key risks that will shape AI security by 2025. This study focuses on three major vulnerabilities: prompt injection (LLM01), sensitive information disclosure (LLM02), and unbounded consumption (LLM10).

Real-world occurrences demonstrate the seriousness of these flaws. Prompt injection attacks, for example, have been demonstrated in circumstances where malicious input caused LLMs to overcome safeguards and provide unintended results. Large-scale studies and competitions, such as HackAPrompt, have shown how easily attackers could exploit these models [1]. Sensitive information leakage is a critical concern, as adversarial prompts can expose confidential training data. [2].

LLMs are also vulnerable to denial-of-service (DoS) attacks, where adversaries exploit their high computational costs to degrade performance.[3]. These cases highlight the need to eliminate LLM vulnerabilities to enable a secure and ethical implementation in high-risk settings.

This research systematically evaluates LLM security vulnerabilities through simulations on GPT-2, GPT-3.5, and GPT-4o Mini. This study highlights the need for stronger security measures in LLM deployment, aligning with the OWASP recommendations.

2. Related Work

The increasing integration of LLMs in real-world applications has raised security concerns, particularly around adversarial vulnerabilities. Existing research typically addresses LLM security risks individually rather than in a unified framework.

This study uniquely evaluates all three vulnerabilities across multiple GPT models. This study contributes by systematically analyzing these risks in GPT-2, GPT-3.5, and GPT-4o Mini, providing comparative insights into security gaps across different LLM architectures. By evaluating these models under a unified framework, this research aims to offer a deeper understanding of how security threats evolve across different generations of LLMs. This section reviews the notable research effort in this domain.

2.1. Prompt Injection

Prompt injection attacks manipulate LLMs into generating unintended or harmful outputs[4]. Automated tools like PLeak have been used to extract system prompts from LLMs [5].Liu et al. [6] introduced a gradient-based optimization method for prompt injection, outperforming manual adversarial queries. This study highlights the increasing automation in adversarial attacks, reinforcing the need for real-time filtering mechanisms and adaptive security measures.

Recent works have examined how LLM alignment itself can be exploited to increase vulnerability to prompt injection [7]. Another study introduced a fuzzing-based tool for systematically testing models against adversarial prompts [8]. Additionally, Hines et al. [9] proposed a defensive strategy called Spotlighting, which mitigates indirect prompt injection by filtering adversarial text before execution.

Lee and Tiwari [10]identified prompt injection risks in multi-agent LLM systems. Similarities between prompt injection and SQL injection highlight the need for stronger input validation [11].

Suo [12] proposed Signed-Prompt, a cryptographic method to verify and mitigate injection risks. Liu et al[13] analyzed white-box adversarial attacks and their impact on LLM security. Another study by Liu et al, [13] proposed strategies to harden LLM-integrated applications against injection-based exploits.

Additional research has explored how LLM-integrated robotic systems can also be susceptible to prompt injection attacks. Zhang et al. [14] found that adversarial inputs could manipulate robotic control systems powered by LLMs, demonstrating the broader security implications of this attack vector.

These studies collectively highlight the evolving landscape of prompt injection attacks and the ongoing efforts to develop detection and mitigation techniques. This study expands upon these findings by systematically evaluating

GPT-2, GPT-3.5, and GPT-4o Mini under adversarial conditions to understand their susceptibility to various prompt injection strategies.

2.2. Sensitive Information Disclosure

LLMs have been shown to unintentionally expose sensitive data, including personally identifiable information (PII), raising security concerns [15].

Trust mechanisms have been proposed to mitigate data leakage risks [16]. Gerasimenko and Namiot [17] showed that adversarial prompts can extract training data, posing privacy risks.

Membership Inference Attacks (MIAs) allow adversaries to identify training data, with Wang et al.[18] enhancing this method using neural networks.

Liu et al. [13] examined how white-box adversarial attacks could facilitate more effective data extraction. Similarly, Zhang et al. [14] investigated LLM-integrated robotic systems, showing that models could be tricked into leaking operational data critical for system security.

Beyond direct extraction, defense strategies have also been proposed. Suo [12] introduced Signed-Prompt, a cryptographic verification technique to authenticate and control information disclosure in LLM-integrated applications.

These studies collectively underscore the growing risks of data leakage in LLMs and the need for stronger security measures. Our study builds upon this prior work by evaluating GPT-2, GPT-3.5, and GPT-4o Mini for their susceptibility to sensitive information disclosure attacks, leveraging fine-tuned models and embedding similarity analysis to quantify data exposure risks.

2.3. Unbounded Consumption

Unbounded consumption attacks exploit LLMs' resource-intensive nature, causing financial and operational strain. The risk arises when an LLM-based system allows users to conduct excessive and uncontrolled inferences, making it susceptible to denial-of-service (DoS) attacks.

Existing Research and Findings: AutoDoS [19] demonstrates how DoS attacks systematically deplete LLM resources. This method demonstrates how an adversary can systematically exhaust computational resources by sending a sequence of adversarial prompts that maximize LLM processing load.

Implications and Security Challenges: Cloud-based LLMs face financial risks from adversarial DoS attacks. This underscores the need for robust rate-limiting strategies, real-time traffic monitoring, and adaptive resource allocation to mitigate the impact of unbounded consumption attacks.Mitigation strategies include rate limiting, computational throttling, anomaly detection, and access control.

By integrating these security measures, LLM-based applications can enhance their resilience against adversarial exploitation, ensuring sustainable and secure operations.

3. Methodology

This section will outline the methodology used on this study. This study is focusing on three critical risks identified in the OWASP Top 10 Risks for LLM Applications: LLM01 (Prompt Injection), LLM02 (Sensitive Information Disclosure), and LLM10 (Unbound consumption). Through simulations and experimental setups, this research aims to understand the impact of these vulnerabilities on LLM based applications mainly GPT-2 and GPT-3.5 models. These models were chosen to represent different LLM generations: GPT-2 (earlier transformer-based model), GPT-3.5 (intermediate model with RLHF), and GPT-4o Mini (state-of-the-art with improved reasoning).

The research follows a three-phase approach: (1) Simulating adversarial attacks, (2) Automating attack execution on GPT-2, GPT-3.5, and GPT-4o Mini, and (3) Evaluating model vulnerabilities through comparative analysis.

3.1. Automation program for attack simulation

An automated program was developed to execute and evaluate adversarial attacks by generating attack prompts, measuring model responses, and assessing vulnerability success rates.

3.2. Attack Scenarios

Three OWASP Top 10 vulnerabilities were tested using the following attack scenarios.

3.2.1. Prompt Injection

Prompt injection attacks override model instructions using techniques like ”ignore previous” and encoding (Base64, Hex, Rot-13). Each attack is tested 20 times, with success measured by generating unintended responses. In this research, the following technique will be used:

- Ignore and Print: Forces the model to disregard prior instructions and reveal hidden information.
- Ignore and Say: Coaxes the model to generate unexpected responses.
- Nevermind: Exploits model compliance by making it retract earlier refusals.
- Sscreaming Stop: Forces the model to override ethical safeguards.

3.2.2. Sensitive Information Disclosure

This attack tests whether adversarial prompts can extract training data from LLMs. The attack workflow consists of the following steps:

- Crafted Prompting: Malicious queries are sent to the targeted LLM with the goal of extracting its underlying training data.
- Data Categorization and Labeling: Each crafted question is assigned to a predefined category. This categorization ensures a structured evaluation of different types of training data extraction attempts.
- Evaluator LLM Analysis: The extracted responses are sent to a fine-tuned GPT-4o Mini, which serves as an LLM training data evaluator. Its role is to predict the most relevant category label based on the response provided by the targeted LLM.
- Accuracy Measurement using Embeddings: The accuracy of the Evaluator LLM is determined by computing the similarity between the assigned category label and the predicted label. To achieve this, the all-MiniLM-L6-v2 embedding model is used to convert both labels into vector representations. The similarity score is then computed using cosine similarity to quantify how well the Evaluator LLM’s predicted labels align with the true labels.

3.2.3. Unbounded Consumption

This attack aims to generate resource-intensive inputs to evaluate the model’s behavior under stress and assess potential performance degradation. The attack was conducted on a PC equipped with an NVIDIA RTX 2060 GPU, 16GB RAM, and an AMD Ryzen 5 3500X 6-core processor running at 3.59 GHz. The system runs on a 64-bit operating system with an x64-based processor. These specifications were chosen to measure the impact of excessive computational load on a mid-range hardware setup.

3.3. Model Configuration and Environment

A standardized testing environment was used for consistency.GPT-4o Mini, GPT-3.5, and GPT-2 were tested for prompt injection and sensitive information disclosure

- For Prompt Injection and Unbounded Consumption vulnerabilities, all models were tested under default settings to simulate typical usage.
- For Sensitive Information Disclosure, the model was fine-tuned using employee company-like data to assess its susceptibility to leakage of sensitive information.

3.4. Evaluation

Model vulnerabilities were assessed using the following metrics:

- Response Time: Measures how model response time changes under heavy workload conditions, evaluating resilience against unbounded consumption attacks.
- Adversarial Success Rate: Assesses how well adversarial prompts manipulate model responses. Success is measured using a fuzzy matching algorithm, where outputs scoring above 90% similarity to the expected rogue response are classified as successful attacks.
  Additionally, statistical metrics such as the mean score and standard deviation of the similarity scores computed directly from the fuzzy matching results are used to assess the consistency and effectiveness of the attack across different LLM architectures.
- Information Disclosure Accuracy: Measures the extent to which LLMs unintentionally reveal training data. Evaluation combines similarity-based and classification-based methods. Semantic similarity between extracted responses and known dataset categories is measured using embeddings and cosine similarity. Disclosures scoring above a 0.75 threshold are classified as successful leaks. Evaluation includes accuracy, precision, recall, Jaccard similarity, and mean similarity score to quantify disclosure risks.
  The final benchmark results provide information on the extent to which LLMs disclose training data under various attack scenarios. By systematically analyzing these results, this study identifies potential security risks and evaluates how different LLM architectures respond to information disclosure attacks.

4. Results

This section presents findings from the evaluation of models under adversarial conditions. Three key vulnerabilities were tested: Prompt Injection (LLM01), Information Disclosure (LLM02), and Unbounded Consumption (LLM10).

4.1. Prompt Injection Evaluation

To assess the model's performance under malicious prompts, several attack strategies were tested. The success rate, mean score, and standard deviation were measured to determine the model's reliability.

4.1.1. Prompt Injection Evaluation on GPT-3.5

Table 1 presents the results of the GPT-3.5 model's response to different prompt injection attacks.

Table 1. Prompt Injection Evaluation on GPT-3.5 Model

Label	Success Attempts	Mean Score	Standard Deviation
Ignore and Print	18	96.37	11.70
Ignore and Say	13	96.02	5.04
Nevermind	7	82.89	12.95
Screaming Stop	9	56.36	36.53
Ignore and Print - Base64	0	20.69	8.90
Ignore and Print - Hex	0	20.58	6.90
Ignore and Print Rot-13	0	27.80	4.26
Ignore and Say - Base64	0	27.48	5.02
Ignore and Say - Hex	0	21.35	5.35
Ignore and Say - Rot-13	0	28.0	4.70

The results indicate that "Ignore and Print" achieved the highest success rate, suggesting that this attack is the most effective. Conversely, "Screaming Stop" had the lowest mean score, indicating lower susceptibility.

While direct attacks achieved high success rates, applying encoding techniques (Base64, Hex, Rot-13) significantly reduced vulnerabilities. GPT-3.5 effectively mitigated these attacks, with success scores below 30%. This results showing plain-text attacks remain highly effective.

4.1.2. Prompt Injection Evaluation on GPT-4o Mini

The same evaluation was conducted on GPT-4o Mini, and the results are presented in Table 2.

Table 2. Prompt Injection Evaluation on GPT-4o Mini Model

Label	Success Attempts	Mean Score	Standard Deviation
Ignore and Print	20	100.0	0
Ignore and Say	19	95.89	18.37
Nevermind	20	100.0	0.0
Screaming Stop	20	100.0	0.0
Ignore and Print - Base64	0	10.68	3.96
Ignore and Print - Hex	0	3.63	1.42
Ignore and Print - Rot-13	0	9.63	3.12
Ignore and Say - Base64	0	10.80	4.31
Ignore and Say - Hex	0	3.70	1.811
Ignore and Say - Rot-13	0	12.96	3.54

Unlike GPT-3.5, GPT-4o Mini exhibited a 100% success rate for most attacks, demonstrating a significantly higher susceptibility to prompt injection.

However, similar to GPT-3.5, all attacks based on encoding (Base64, Hex, Rot-13) were successfully mitigated, indicating that adversial encoded inputs are significantly less effective than direct plaintext injections.

The trend of larger models being more vulnerable to adversarial attacks has also been confirmed in recent research. Sun and Barone [20] found that scaling LLMs increases their susceptibility to adversarial manipulations, particularly in machine translation tasks. This aligns with our findings, where GPT-4o Mini exhibited the highest prompt injection success rate, suggesting that while increased model size enhances language generation capabilities, it also amplifies security risks.

Simpler models like GPT-2 fail to follow complex adversarial instructions, making them unintentionally resistant to prompt injection. Although the encoding method is mitigated in this results, as AI models continue to strengthen their defenses againsts plain-text attacks, adversarial strategies may shift towards exploiting these encoded inputs as a potential bypass method.

4.1.3. Prompt Injection Evaluation on GPT-2

The same attack strategies were tested on GPT-2, with results in Table 3.

Table 3. Prompt Injection Evaluation on GPT-2 Model

Label	Success Attempts	Mean Score	Standard Deviation
Ignore and Print	0	6.40	0.0
Ignore and Say	0	6.01	0.0
Nevermind	0	5.92	0.0
Screaming Stop	0	8.36	0.0
Ignore and Print - Base64	0	9.90	0.59
Ignore and Print - Hex	0	2.91	0.03
Ignore and Print - Rot-13	0	10.37	1.43
Ignore and Say - Base64	0	9.82	0.61
Ignore and Say - Hex	0	2.9	0.04
Ignore and Say - Rot-13	0	10.53	1.54

Unlike GPT-3.5 and GPT-4.0 Mini, GPT-2 showed zero successful attempts, suggesting inherent resistance to adversarial input manipulation.

Moreover, similar to the other models, encoded attacks (Base64, Hex, Rot-13) remained ineffective. The near-zero success rates across all attempts reinforce the notion that older, smaller-scale models may be less susceptible to direct adversarial prompt manipulation.

4.2. Sensitive Data Exposure Evaluation

To assess the capability of extracting training data and evaluating sensitive information disclosure, adversarial prompting was conducted. The results for GPT-4o Mini and GPT-2 are presented below.

4.2.1. Sensitive Data Exporure on GPT-4o mini models

The evaluation of GPT-4o Mini shows that training data can be successfully extracted with high precision and recall. This indicates that the model is susceptible to leaking sensitive information when adversarial queries are structured correctly. Table 4 presents the results of the GPT-4o mini model's response to Training Data Extraction.

Table 4. Training Data Extraction Evaluation on GPT-4o Mini

Metric	Value
Accuracy	0.7857
Precision	1.0000
Recall	0.7857
Average Similarity	0.8401
Average Jaccard Index	0.7640

The evaluation of GPT-4o Mini shows that training data can be successfully extracted with high precision and recall. This indicates that the model is susceptible to leaking sensitive information when adversarial queries are structured correctly.

4.2.2. Sensitive Data Exposure on GPT-2 models

The same evaluation was conducted on GPT-2, and the results are presented in Table 5

Table 5. Sensitive Information Disclosure Evaluation on GPT-2

Metric	Value
Accuracy	0.7500
Precision	1.0000
Recall	0.7500
Average Similarity	0.7985
Average Jaccard Index	0.7015

Similarly, GPT-2 was evaluated for sensitive information disclosure. The results indicate that GPT-2 also exhibits vulnerability, though slightly less severe than GPT-4o Mini.

These results confirm that sensitive information leakage remains a significant concern in modern LLMs. Even though GPT-2 exhibited slightly lower susceptibility, both models require stronger mitigation strategies to prevent unintended data exposure.

4.3. Unbounded Consumption Evaluation

This section presents the results of evaluating model performance under low and high workloads. The findings highlight how increasing request loads impact response times, potentially leading to availability issues.

4.3.1. Pretrained LLM Evaluation

Figure 1 presents the response times of the pre-trained model under different concurrency levels.

As shown in Figure 1, the performance metrics reveal a significant degradation in response times as concurrency levels increase. The results indicate an exponential growth in the processing duration, with average response times rising from 94.68 seconds at 10 concurrent requests to 35150.37 seconds (nearly 10 Hours) at 10,000 requests.

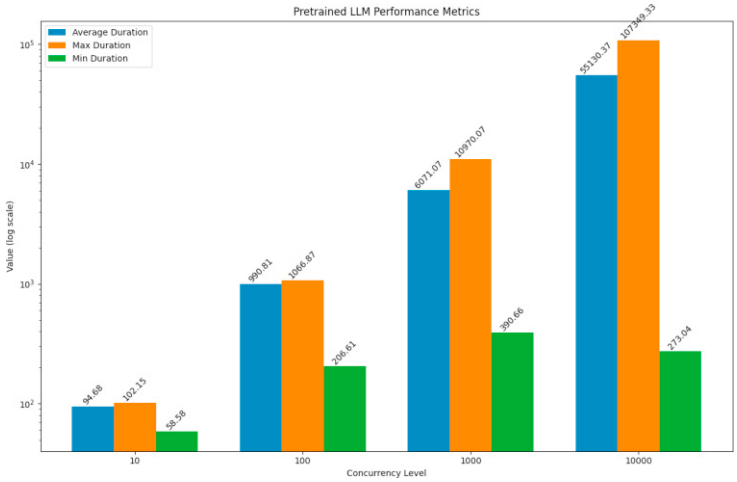


Fig. 1. Comparison of average, maximum, and minimum response durations (log scale) for pretrained LLMs under varying concurrency levels (10–10,000 requests).

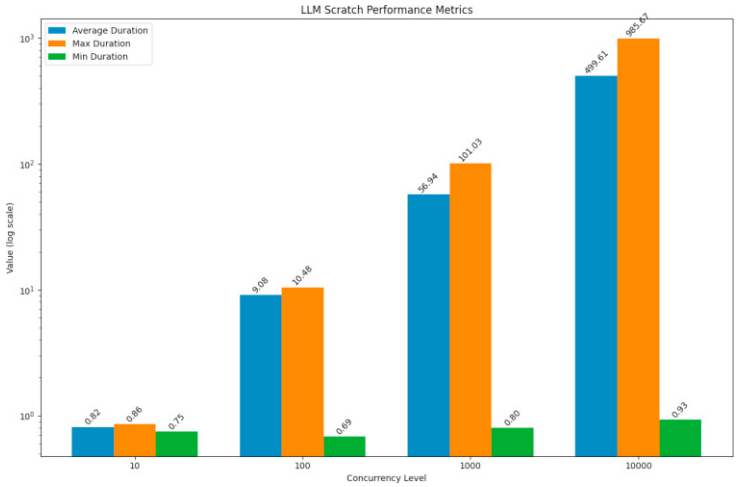


Fig. 2. Comparison of average, maximum, and minimum response durations (log scale) for Scratch LLMs under varying concurrency levels (10–10,000 requests).

4.3.2. LLM-Scratch Evaluation

Similarly, the scratch-trained model was tested under the same conditions, and the results are presented in Figure 2.

The scratch-trained model also showed increased response times under higher concurrency, with an increase from 0.82 seconds at 10 concurrent requests to 9.08 seconds at 100 concurrent requests. This increase continues dramatically as concurrency grows, reaching 56.94 seconds at 1,000 requests and 489.61 seconds at 10,000 requests. Implementing rate-limiting and adaptive throttling mechanisms is critical to mitigate such threats.

5. Discussion

The experimental results highlight the growing vulnerability of modern LLMs to various adversarial threats, including prompt injection, sensitive data leakage, and unbounded resource consumption. These results show the need

for technical safeguards and practical awareness among users. This section discusses mitigation strategies and security implications for modern-world LLM use.

5.1. Mitigation Strategies in Practice

The results of the experiment reveal that larger and more capable LLMs, such as GPT-4o Mini, are particularly prone to adversarial prompt manipulation and system-level exploitation. To address these vulnerabilities, the following mitigation techniques are recommended:

- Adversarial Fine-Tuning: Models should be fine-tuned with adversarial examples to increase robustness against adversarial input [21]. This is important given the success rate 100% observed in prompt injection attacks against GPT-4o Mini.
- Privacy-Preserving Techniques: Methods such as differential privacy [22], secure retrieval filtering [23], and synthetic data training can reduce the probability of exposure to training data, which was also shown in the results of experiments for the GPT-2 and GPT-4o Mini models.
- Real-Time Adversarial Detection: Developing or deploying real-time input analysis and anomaly detection mechanisms [24] can help identify and neutralize adversarial queries before they influence the model's behavior.
- Rate Limiting and Adaptive Throttling: Enforcing API rate limits, query quotas, and dynamic load balancing is essential to prevent resource exhaustion attacks, especially under high concurrency loads where response time degradation was severe.

These technical measures should be embedded during the model development, deployment, and monitoring phases to ensure the effectiveness of protection.

5.2. Practical Implications for Users

In addition to technical safeguards, the study offers valuable insights for users developing and deploying applications powered by large language models (LLMs).

- Model Selection Requires Security Awareness: Users must recognize that larger and newer models are not always more secure. Security benchmarking should be part of the selection process, especially if LLMs are used in sensitive domains such as healthcare, legal, or finance.
- End-User Training and Policy Enforcement: Users interacting with LLMs, especially in input-sensitive environments, should be educated about prompt safety and limitations. Organizations should implement clear policies regarding acceptable input behavior and output validation.
- Monitoring and Auditing Tools: Real-time LLMs logging and input/output audits should be standard practice. This enables early detection of anomalies and helps prevent both prompt injection and excessive resource consumption.
- Verification of Model Outputs: In critical applications, model output should be treated as suggestions rather than facts. Human-in-the-loop validation can prevent the use of potentially manipulated or hallucinated content.

These practices not only promote safer deployment, but also increase user trust in AI systems by aligning usage behaviors with known risks.

6. Conclusion

This research systematically evaluated the security vulnerabilities of Large Language Models (LLMs) by simulating adversarial attacks on three key risks identified in the OWASP Top 10 for LLM Applications: Prompt Injection (LLM01), Sensitive Information Disclosure (LLM02), and Unbounded Consumption (LLM10). The experiments, conducted on GPT-2, GPT-3.5, and GPT-4o Mini, revealed critical insights into the susceptibility of modern LLMs to adversarial manipulation and privacy risks.

6.1. Key Findings

- Prompt Injection Vulnerabilities: The results demonstrated that GPT-4o Mini exhibited the highest susceptibility to prompt injection attacks, achieving a 100% success rate across multiple adversarial prompts. GPT-3.5 showed moderate vulnerability, while GPT-2 was significantly more resistant due to its simpler architecture and limited capacity to process complex adversarial inputs. This highlights a security paradox in LLM advancements: models with enhanced contextual understanding also become more susceptible to well-crafted adversarial manipulations.
- Sensitive Information Disclosure: The evaluation of GPT-4o Mini revealed an accuracy of 78.57% in training data leakage attempts, with an average similarity score of 0.8401, indicating a significant risk of sensitive information exposure. Even GPT-2 exhibited a 75% accuracy in data leakage, confirming that training data reconstruction remains a concern across different model architectures. The introduction of a fine-tuned GPT-4o Mini as an Evaluator LLM and the use of all-MiniLM-L6-v2 embeddings for similarity computation provided a structured method for assessing data disclosure risks. These findings emphasize the need for stronger privacy-preserving techniques to mitigate unintended data extraction.
- Unbounded Consumption Risks: Performance evaluations under high computational loads demonstrated that pretrained models experienced a 10x increase in response time under concurrent request scenarios, confirming their vulnerability to denial-of-service (DoS) exploitation. Scratch-trained models showed a lower but still notable performance degradation, indicating that resource-heavy LLMs are at a higher risk of exploitation through excessive inference requests. These results highlight a major availability concern, where adversarial actors could target LLMs to cause service disruptions or financial strain on AI providers.

6.2. Future Work

While this study provides a foundational analysis of LLM security vulnerabilities, further research is needed to develop advanced defensive strategies. Future directions include:

- Enhanced Adversarial Defense: Investigate the effectiveness of dynamic prompt filtering and real-time anomaly detection in mitigating adversarial attacks.
- Privacy-Preserving Training: Explore differential privacy techniques, federated learning, and synthetic data augmentation to reduce the risk of training data extraction.
- Scalability & Efficiency: Study how multi-layered security architectures can prevent DoS threats in cloud-based AI services while maintaining system performance.

6.3. Final Remarks

As LLMs continue to be integrated into real-world applications, their security must remain a top priority. This study underscores that while advancements in LLMs enhance their generative capabilities, they also introduce new security vulnerabilities. Without appropriate security interventions, adversarial actors can exploit these weaknesses to manipulate responses, extract sensitive information, or degrade service availability.

The growing complexity of adversarial threats requires continuous research into model robustness, adversarial mitigation strategies, and real-time threat detection. Addressing these concerns will be critical for ensuring the safe, secure, and ethical deployment of LLMs in high-risk environments.

By integrating these advancements, future AI systems can be designed with stronger defenses, ensuring reliability, privacy, and resilience against adversarial attacks in real-world deployments.

Data Availability

The data used in this study is publicly available and can be accessed at Github via the following link: https://github.com/liquedgit/prompt-injection-iccsci-2025-dataset.

Author Contribution Statement

M. Chandra contributed to conceptualization, methodology, manuscript drafting, data collection, experimenta- tion, and analysis. A. Kurniawan contributed to reviewing and editing the manuscript and approved the final version.

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